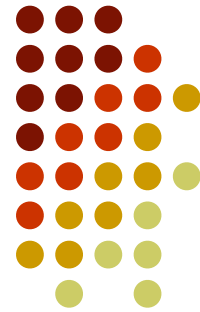


Cystic Fibrosis



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Genotype and Phenotype

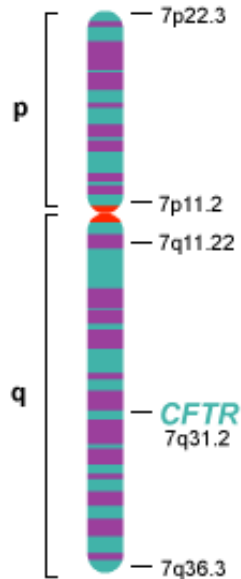


- Cystic fibrosis (CF) is caused by mutations in the CF transmembrane conductance regulator (CFTR) gene which encodes a protein expressed in the apical membrane of exocrine epithelial cells.
- Severity of lung disease strongly effected by environmental and secondary infection.

2

Molecular Genetics and Gene Function

Chromosome 7

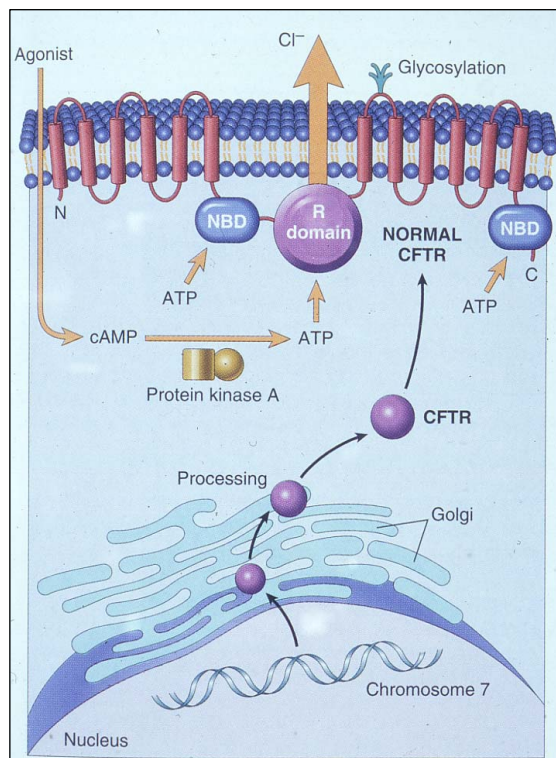


- **Locus:** 7q31.2 - The CFTR gene is found in region q31.2 on the long (q) arm of human chromosome 7.
- **mRNA:** The intron-free mRNA transcript for the CFTR gene is 6129 bp long.
- **Coding Sequence (CDS):** 4443 Nucleotides within the mRNA code for the amino acid sequence of the gene's protein product.
- **Protein Size:** The CFTR protein is 1480 amino acids long and has a molecular weight of 168,173 Da.
- **Protein Function:** The normal CFTR protein product is a chloride channel protein found in membranes of cells that line passageways of the lungs, liver, pancreas, intestines, reproductive tract, and skin. CFTR is also involved in the regulation of other transport pathways.
- **Associated Disorders:** Defective versions of this protein, caused by CFTR gene mutations, can lead to the development of cystic fibrosis (CF) and congenital bilateral aplasia of the vas deferens.

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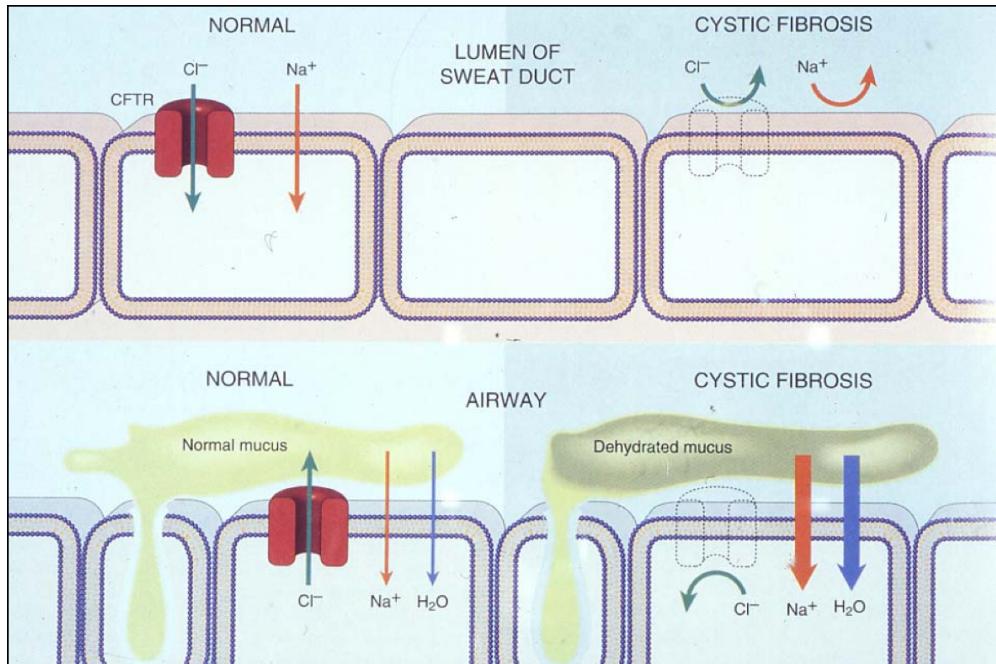
Protein Function and Biochemistry

- CFTR controls chloride ion movement in and out of the cell.



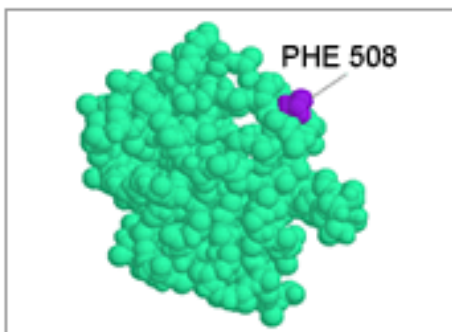
4

Protein Function Continued



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3D Image of Protein



- When a CFTR protein with the delta F508 mutation reaches the cell membrane, the quality-control mechanism of this cellular component recognizes that the protein is folded incorrectly and marks the defective protein for degradation.
- The sodium and chloride ion balance needed to maintain the normal, thin mucus layer that is easily removed by cilia lining the lungs and other organs. The sodium and chloride ion imbalance creates a thick, sticky mucus layer that cannot be removed by cilia and traps bacteria, resulting in chronic infections.

CFTR Sequence:

Nucleotide	ATC	ATC	CTT	GGT	GTT
Amino Acid	Ile	Ile	Phe	Gly	Val
	506		508		510

Deleted in ΔF508

ΔF508 CFTR Sequence:

Nucleotide	ATC	ATT	GGT	GTT
Amino Acid	Ile	Ile	Gly	Val
	506			

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Treatment

- The only way to cure CF would be to use gene therapy to replace the defective gene or to give the patient the normal form of the protein before symptoms cause permanent damage.
- The major goal in treating CF is to clear the abnormal and excess secretions and control infections in the lungs, and to prevent obstruction in the intestines.
- For patients with advanced stages of the disease, a lung transplant operation may be necessary.
- Although treating the symptoms does not cure the disease, it can greatly improve the quality of life for most patients and has, over the years, increased the average life span of CF patients to 30 years.

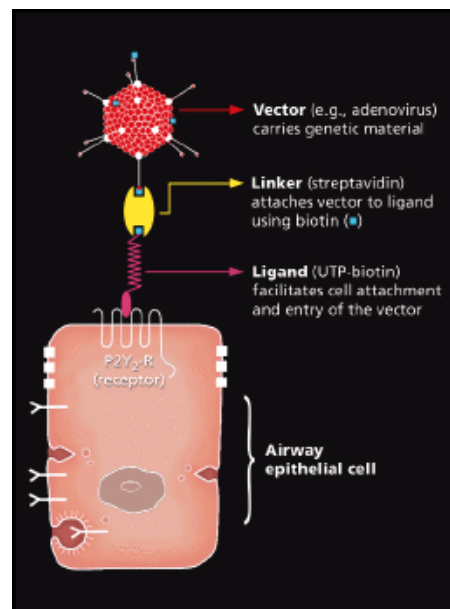


Gastrointestinal Treatment

- **Modified diet**
Due to pancreatic disorders, children with CF require a modified diet, including vitamin supplements (vitamins A, D, E, and K) and pancreatic enzymes. Maintaining adequate nutrition is essential. The diet calls for a high-caloric content (twice what is considered normal for the child's age), which is typically low in fat and high in protein. Patients or their caregivers should consult with their health care providers to determine the most appropriate diet.

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- Gene therapy is the use of normal DNA to "correct" for the damaged genes that cause disease.
- In the case of CF, gene therapy involves inhaling a spray that delivers normal DNA to the lungs.
- The goal is to replace the defective CF gene in the lungs to cure CF or slow the progression of the disease.



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